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A last-mile drone-assisted one-to-one pickup and delivery problem with multi-visit drone trips

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ABSTRACT

Online retailers and food delivery platforms are exploring last-mile package and food delivery using drones to reduce delivery time and improve customer experience. We investigate a novel and pioneering problem, the one-to-one pickup and delivery problem with multi-trucks and multi-visit drone (OPDP-MTMV). This problem involves multiple trucks, each of which is equipped with a single drone for pickup and delivery services. The drones are capable of carrying multiple packages per drone trip. Pickup and delivery requests can be served by either a truck or a drone, subject to the one-to-one precedence constraints of the requests, the capacity constraints of the truck and the drone, and the endurance constraints of the drone. The energy consumption of a drone is based on the flight time and the varying payload along the multi-visit drone trip. The objective is to minimize the sum of the fixed cost and the variable duration cost. We formulate this problem using a mathematical model and propose an iterated local search (ILS) algorithm based on the distinct features of the problem. Finally, we conduct comprehensive computational experiments using test instances of various sizes to analyze the characteristics of the solution, the applicability of the algorithm, and the impact of different drone configurations.

1. Introduction

Drone delivery operations have gained wide attention recently given their potential to save time and cost while reducing environmental impact (Campbell et al., 2017). Companies such as Amazon, Google, Alibaba, JD, Wal-Mart, DHL, UPS, and SF Express, have conducted various trial runs in this regard (Macrina et al., 2020). Concurrent research studies have focused on similar applications with different operational constraints and regulatory rules (Murray and Chu, 2015).

Among such applications, food delivery is one of the most prominent (Kelso, 2019) because drone delivery could potentially improve the delivery speed of perishable food items. This, in turn, would lead to higher customer satisfaction, improved operational efficiencies, and lower overall costs in the long run. In a trial run conducted in August 2020, FoodPanda, a food delivery service provider, deployed a drone to deliver a package of food from mainland Singapore to a ship located 3 km offshore (Fig. 1).

Karak and Abdelghany (2019) studied a truck–drone cooperative pickup and delivery services problem with a simplified drone endurance model and without precedence constraints, demonstrating the effective cost reduction achieved with the cooperative model. Recently, a German aircraft developer named Wingcopter released a new drone that can make three delivery drop-offs on a single flight (Fig. 2). Such developments expand the opportunities for exploring drone delivery applications with multiple packages per drone trip.

This study investigates a novel and pioneering last-mile delivery problem named the one-to-one pickup and delivery problem with multi-truck and multi-visit drone (OPDP-MTMV), which deploys multiple truck–drone pairs to serve customers. Each customer order requires a package to be transported from one location to another location. Both the trucks and the drones can perform pickup and delivery operations, while the drone can be launched from the truck at any customer location when the truck is stationary. At the launch node, the drone will be loaded with packages for delivery, whereas at the retrieval node, the packages collected by the drone will be unloaded onto the truck. Both the trucks and the drones are constrained by their respective maximum weight capacities. An energy consumption constraint is also enforced for each drone based on both flight time and total weight carried by

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Fig. 1. Drone delivery trial by Foodpanda (Abdullah, 2020).

Fig. 2. Wingcopter 198 (Aria, 2021).

the drone. The OPDP-MTMV aims to minimize both the fixed cost of the truck–drone pairs deployed and the variable duration cost of all the trips. Fig. 3 depicts an OPDP-MTMV solution with two truck–drone pairs for 10 requests. The locations i^+ and i^- represent the pickup and delivery nodes, respectively, for customer i. Truck 1 visits seven nodes before it returns to the depot, while Drone 1 performs three drone trips and serves a total of five nodes.

The OPDP-MTMV problem opens up new research opportunities and motivates future trial runs and deployment with its unique and interesting features: (1) Pickup and delivery of a request can be handled by either a truck or its paired drone as long as the precedence constraint is observed (Fig. 4 shows three possible hybrid service models for the problem); (2) Delivery of the package is not required immediately after the pickup operation (For example, Package 1 is picked up by Drone 1 and delivered by Truck 1 at a later time in the example depicted in Fig. 3); (3) A drone can serve multiple locations per flight (Wang et al., 2019; Gonzalez-R et al., 2020; Luo et al., 2021), which raises an interesting and challenging problem for the weight and flight-time-based endurance model for the drone; (4) The truck routes and their associated drone trips must be synchronized to enable the replacement of the drone battery, the exchanges of packages, and the assessment of drone energy consumption. These features establish the OPDP-MTMV as a challenging problem to model and solve. Our contributions include: (1) A novel and interesting variant of the last-mile delivery problem using trucks and drones; (2) A mathematical programming model based on the decomposition of the combined route into partial routes; (3) A special endurance model that considers the total weight carried, the flight time, and the hovering time of the drone (the quadratic energy consumption function is approximated by two linear terms in the model); (4) An iterated local search (ILS) metaheuristic with a three-stage evaluation method, tailored local search operators, and six perturbation operators; (5) A set of test instances with various drone settings and comprehensive computational experiments to evaluate the effectiveness of the model.

The remainder of the paper is organized as follows. Section 2 provides a review of the relevant literature. Section 3 presents the problem in detail. Section 4 describes the heuristic algorithm in detail. The experimental studies are provided in Section 5, including analysis of the results. Finally, Section 6 presents our conclusions and a discussion of future research.

2. Related work

Otto et al. (2018), Macrina et al. (2020) and Li et al. (2021) provided detailed reviews on drone routing problems by categorizing them into independent drone delivery operations and cooperative vehicle–drone delivery operations. Furthermore, Otto et al. (2018) differentiated problems under the cooperative truck–drone delivery category based on whether the truck only served as a base for the drones or if both trucks and drones were allowed to serve customers. The OPDP-MTMV problem corresponds to the latter option with the synchronization constraint that requires the trucks and their equipped drones to wait for each other at the rendezvous point. The synchronization constraint also exists in those vehicle routing problems with synchronization constraints (Drexl, 2012) . Lastly, the OPDP-MTMV considers one-to-one pickup and delivery requests yielding to precedence constraints, a topic which has been widely studied in the research literature (Cordeau et al., 2008).

2.1. The basic truck–drone routing problem

Murray and Chu (2015) first proposed the deployment of both a truck and a drone together for package delivery in logistics distribution, naming the problem the *Flying Sidekick Traveling Salesman Problem* (FSTSP). Agatz et al. (2018) proposed a similar problem called the *Traveling Salesman Problem with Drones* (TSP-D), which was solved using a truck-first-drone-second heuristic and a dynamic programming algorithm. The main difference between the FSTSP and the TSP-D is that the TSP-D model restricts drone trips from starting and ending at the depot even though it allows drone trips that start and end at the same customer node. Such drone trips are commonly known as drone *loop* operations (Roberti and Ruthmair, 2021). Agatz et al. (2018) also allowed trucks to revisit nodes that had already been served. However, loop operations and the revisit feature have not been widely adopted in the literature.

In the previous series of articles represented by FSTSP, the flight process of the drone was constructed with a three-dimensional decision variable. With such a construction method, it is not easy to calculate the specific time (Das et al., 2020; Kuo et al., 2022) when the drone starts to serve customers if there are multiple loops in a node. Therefore, another type of arc-based modeling method, named *"extended graph"* (Boccia et al., 2021), is proposed to combine the flight of the drone and the process of being moved in a truck to form a continuous route, where the decision variable is two-dimensional and uses similar routing constraints as a conventional *vehicle routing problem* (VRP). In this case, the route of the drone needs to remain a simple tour construction (Poikonen et al., 2019), that is, subtours or repeated visits to the arcs and nodes are not allowed, so as to avoid conflicts when calculating the times of arrival at customer nodes.

In various studies, no significant distinction has been made between the FSTSP and the TSP-D. In Yurek and Ozmutlu (2018), the authors directly investigated the common characteristics of the FSTSP and TSP-D as the same type of problem. Similarly, in several review articles (Macrina et al., 2020; Li et al., 2021; Moshref-Javadi and Winkenbach, 2021), these two types of problems are usually grouped under the same category. Certain variations of these two types of problems are also summarized and compared in these reviews.

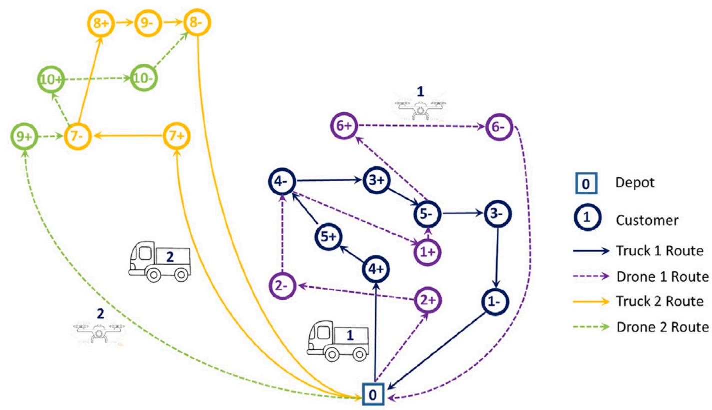


Fig. 3. An example of the OPDP-MTMV problem.

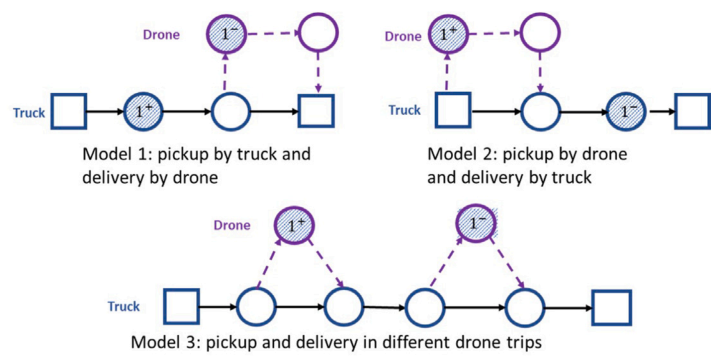


Fig. 4. Three hybrid service models.

The *Vehicle Routing Problem with Drones* (VRP-D) problem is an important extension of the TSP-D, which deploys multiple vehicles equipped with drones to perform deliveries. Wang et al. (2017) provided worst-case analyses of the impact of the number of drones per truck and the speed of the drones on the maximum savings in comparison with the TSP solution costs. Poikonen et al. (2017) extended the discussion in Wang et al. (2017) by considering the drone's battery life, cost metrics, and the fixed cost of deploying drones. A mixed integer linear program was proposed by Schermer et al. (2019b), exploiting the VRP-D solution structure. Another mixed integer program with a time limitation, where minimizing cost was the objective function, was introduced by Sacramento et al. (2019). An adaptive large neighborhood search metaheuristic was proposed to solve instances with at most 200 customers.

Several other studies considered various aspects of the VRP-D. Di Puglia Pugliese and Guerriero (2017) introduced a VRP-D variant with a customer time window. These authors' subsequent study (Pugliese et al., 2020) indicated that the VRP-D produced lower CO_2 emissions compared with the classical VRP. Schermer et al. (2019a) extended the VRP-D by introducing the en-route operator, which allows the drones to be retrieved not only in customer nodes but also in the edge traveled by the truck. Gu et al. (2022) proposed a general *Vehicle Routing Problem with Drones and Multiple Visits* (VRPD-MV) where the drones are capable of serving multiple customers per trip. A variable neighborhood descent procedure (ILS-VND) and an efficient hierarchical solution evaluation method with the time complexity reduced to $O(1)$ are proposed to solve instances within 200 customers.

All the studies cited above considered only a single customer visit per drone trip. As summarized in Macrina et al. (2020), most studies of the TSP-D and VRP-D enforced a distance or flight time constraint on the drone trip, assuming a constant time for battery swap or charge. Whereas almost all studies enforced truck capacity constraints, only a few considered capacity constraints for drones.

2.2. Extension with multi-visits per drone flight

Although multiple visits per drone trip are commonly seen in surveillance and reconnaissance applications (Campbell et al., 2018; Luo et al., 2017, 2018; Liu et al., 2019), such a setting is rarely investigated in combined truck–drone routing problems. Wang and Sheu (2019) studied the VRP-D including multi-visits, which limits drone routes by the maximum flying time and the maximum drone payload. Kitjacharoenchai et al. (2020) incorporated both multi-visits and multiple drones per truck into the *Two-Echelon Vehicle Routing Problem with Drones* (2EVRPD) with a maximum flight time constraint for each drone. Poikonen and Golden (2020) proposed a similar problem with multi-visits and multi-drones per truck, called the k-*Multi-Visit Drone Routing Problem* (k-MVDRP). The k-MVDRP adopts an energy consumption model based on the payload and the travel distance as proposed in Stolaroff et al. (2018). The k-MVDRP restricts drone trips with the same launch node to end at the same retrieval node. In addition, the truck must travel directly from the launch node to the retrieval node, which simplifies the synchronization constraint for the problem. Gonzalez-R et al. (2020) proposed a multi-visit truck–drone team logistics problem that adopts a travel-distance-based endurance model for the drone.

Recently, Luo et al. (2021) proposed the *Multi-Visit Traveling Salesman Problem with Multiple Drones* (MTSP-MD), which adopts an endurance model based on the weight of the drone itself, its payload, as well as the flight time. Specifically, the MTSP-MD considers energy consumption during the hovering time when a drone is waiting for the truck to arrive. Our OPDP-MTMV problem adopts the same endurance model as the MTSP-MD. However, the drone hovering scenario must be carefully evaluated because the drone could be carrying collected packages while waiting for the truck. The packages on board the drone change the energy consumption during the hovering.

2.3. Pickup and delivery with drones

Pickup-and-Delivery Problems (PDPs) constitute an important family of VRP problems that involve the transportation of goods or passengers from various origins to various destinations.

In this section, we focus on PDP problems with a one-to-one precedence constraint for the request; that is, the pickup demand from one location must be delivered to another specified location (Cordeau et al., 2008). Such problems are commonly investigated in urban courier services (Soysal et al., 2018), passenger transportation (Cordeau, 2006), food delivery (Ulmer et al., 2021), and maritime shipping (Korsvik et al., 2011).

The use of drones in the PDP has yet to be studied extensively. Ham (2018) extended the *Parallel Drone Scheduling Traveling Salesman Problem* (PDSTSP) (proposed by Murray and Chu (2015)) with both pickup and delivery operations to be performed by the drone.

The trucks and drones work in parallel and independent of each other in the PDSTSP, which requires no synchronization of the retrieval process. A more complex synchronization constraint is enforced in the *Hybrid Truck–Drone Routing Problem* (HVDRP) proposed by Karak and Abdelghany (2019), where multi-visit drones are mounted on the trucks (trucks cannot serve customers) to serve both pickup and delivery customers but without precedence constraints.

Two separate strategies are proposed for drone dispatching and collection: (1) Trucks adopting the dispatch-wait-collect strategy wait at the same location to retrieve the drones after launch operations; (2) Trucks adopting the dispatch-move-collect strategy can retrieve any drone, regardless of the truck from which it was launched.

However, adopting these two strategies for our problem may be inefficient or even infeasible. On the one hand, the dispatch-wait-collect strategy is usually adopted when the locations where trucks can stop are limited (Luo et al., 2018). In the context of drone multi-visits, it is relatively more efficient to allow trucks to continue serving customers after launching the drone if the truck is able to perform launch or retrieve operations at all nodes.

On the other hand, the dispatch-move-collect strategy is likely to cause package confusion in one-to-one PDP scenarios and makes problem modeling and problem-solving difficult to implement.

Therefore, while we acknowledge the applicability of both strategies, for the purpose of this study, we assumed that trucks were neither allowed to remain at a location to retrieve drones nor to retrieve drones deployed from other trucks.

3. Problem statement and mathematical model

Notations for the truck and the drone are differentiated by superscripts “G” and “U”, respectively, where “G” represents the *Ground Vehicle* (truck) and “U” represents the *Unmanned Aerial Vehicle* (UAV or drone).

3.1. Notations and problem description

The OPDP-MTMV can be formally defined over the graph (V, E), with V as the node set and E as the arc set. Formally, $V = P \cup D \cup \{0\}$, where 0 represents the depot, and $P = \{1, \dots, n\}$ and $D = \{n+1, \dots, 2n\}$ represent the pickup and delivery nodes, respectively. Each customer $i \in P$ requires a package of weight w_i to be picked up and delivered to location $i + n \in D$. Hence, $w_{i+n} = -w_i \; \forall i \in P$. Note that i^+ and i^- will also be used to represent the pickup node and delivery node of package i, respectively. We use N to represent the set of all customer nodes, i.e, $N = P \cup D$. Formally, E is defined as $\{(i,j)|\ i,j \in V,\ i \neq j,\ i \neq j+n\}$.

A fleet of trucks K, each equipped with a single delivery drone, is deployed to serve all customers. A truck has a maximum weight capacity Q^G and a maximum number of allowed working hours (t_{max}), whereas a drone is restricted by the maximum payload capacity Q^U and the maximum battery energy. We define the **total weight carried by the truck** as the self-weight of the drone on the truck, if it is on board, and the combined weight of all packages on board, including the weight of packages that are loaded on the drone and other packages that are on the truck but not loaded. For clarity, the **total weight of the drone** (when deployed) refers to the self-weight of the drone added to the sum of weights of all loaded packages on the drone, which we refer to as **the drone's payload**. Each customer $i \in N$ must be served once only, by either a drone or a truck, and if a truck visits any customer, the customer should be serviced directly by the truck. A service time of s_i^U or s_i^G is required for a drone or a truck, respectively, to service the pickup or delivery node of customer i. The travel times along arc $(i,j) \in E$ for the truck and the drone are represented by t_{ij}^G and t_{ij}^U, respectively.

When a drone is not used, it will be transported on the truck to the next node. In the OPDP-MTMV, we assume that a drone trip cannot end at the node from where it originated. This is a reasonable assumption since the combination of the drone's higher speed and concurrent travel for the truck and its deployed drone is likely to result in improved delivery efficiency, as trucks could use the drone's airborne time to serve other customers. This assumption was discussed in Liu et al. (2019), whose experiments suggested that efficiencies were increased by disallowing drone loops. The hovering operation for a drone occurs when it arrives at the retrieval node ahead of the truck's arrival. For safety reasons, the drone cannot land on the ground and it continues to consume energy while waiting to be retrieved by the truck.

A truck might arrive at the delivery node of a request while the drone carrying the correct package is still on the way. Therefore, the OPDP-MTMV enforces the following ordering policy at a truck node: (1) The truck arrives; (2) The drone is retrieved by the truck when there is a retrieval operation; (3) The customer is served by the truck/driver; (4) The truck departs from the customer node at the same time as the launched drone if a launch operation is performed.

3.2. Drone energy consumption model

A drone trip is restricted by a maximum payload weight (Q^U) as well as a maximum battery capacity (θ). The energy consumption rate of the drone is directly related to the sum of its self-weight and the weight of the payload, and to the flight time (Zhang et al., 2021; Liu et al., 2020; Luo et al., 2021). Let $f_1(t,w)$ and $f_2(t,w)$ represent the energy consumption during flight time and during customer service time, respectively, where t_{ij}^U represents the flight time between nodes i and j, s_i^U represents the service time at customer i, and w represents the total weight, including both the total payload of the drone and its self-weight. Similarly, let $f_3(t,w)$ be the energy consumption when a drone hovers and waits for t time units with a total weight w. $f_1(t,w)$ and $f_2(t,w)$ are defined below, based on the model in (Luo et al., 2021):

$$f_1(t,w) = \alpha \times \left(w^U + w_i + w_i^U\right) \times t_{ij}^U, \tag{a1}$$

$$f_2(t,w) = \alpha \times \left(w^U + w_i + w_i^U\right) \times s_i^U, \tag{a2}$$

where w_i^U denotes the total payload of the drone when it departs node i, and α represents the energy consumption rate per unit weight, per time unit. Therefore, when calculating the total power consumed at the current node i, the weight of the package at the current node w_i should be included. During hovering, both t and w in $f_3(t,w)$ are unknown decision variables and $f_3(t,w) = t \times w$ becomes a quadratic term. In this study, $f_3(t,w)$ was approximated by two linear terms as follows:

$$f_3(t,w) = \alpha \times \left(\beta_1 \times w + \beta_2 \times t\right) \tag{a3}$$

where β_1 and β_2 are two fixed parameters.[2]

[2] The values of the parameters were set as $\alpha = 1$, $\beta_1 = 5$, and $\beta_2 = 1$.

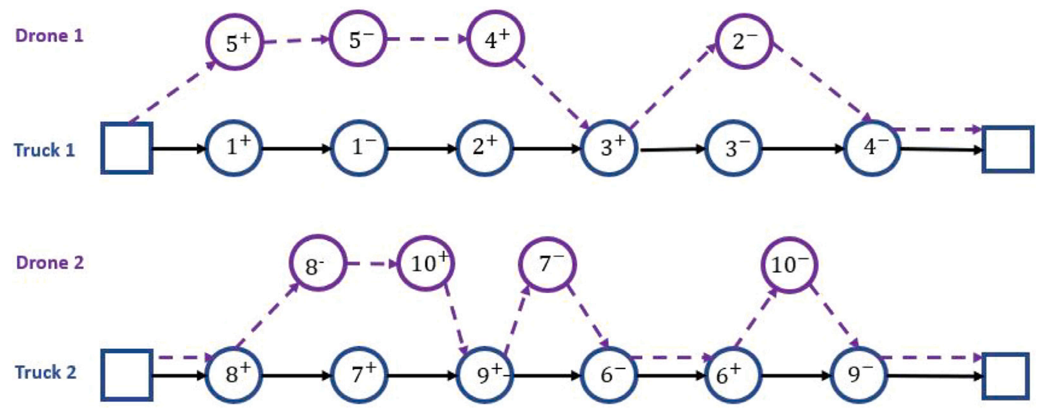


Fig. 5. Schematic diagram of extended graph.

3.3. Modeling approach

The delivery approach associated with the OPDP-MTMV as described in the previous section is illustrated in Fig. 5. This figure involves two types of nodes and arcs indicated with different colors. The customers served by the truck and by the drone are colored blue and purple, respectively. In accordance with the methods of Roberti and Ruthmair (2021) and Boccia et al. (2021), we used *Extended Graph* to describe the routes of the trucks and drones. The truck routes are represented by the solid arrows, while the drone routes are represented by the dashed arrows. In the *Extended Graph* representation, each drone route is described in a simple graph (no repeated edges or loops). The drone arcs are displayed alongside the truck routes when the drone is loaded and transferred by the truck, such as arc $4^- \longrightarrow depot$ in Drone 1's route, and $depot \longrightarrow 8^+$ in Drone 2's route. The objective was to minimize the total operational costs for the truck–drone pairs, which included the total duration cost of all combined routes and the fixed costs for the truck–drone pairs, with λ_1 and λ_2 as numerical parameters that quantify the respective costs.[3]

Six important decisions need to be made simultaneously to obtain the solution: (1) Determine customer assignments and prevent the package delivered by other truck–drone combination; (2) Determine the customers visiting order of trucks and drones; (3) Determine the assignment of packages on trucks and drones and ensure that capacity limits are not exceeded; (4) Determine the launching and retrieving nodes; (5) Calculate the time of arrival and departure at each node; (6) Calculate the amount of energy consumed during each flight and ensure that the drone can return to the truck before the battery runs out.

Most of the above decisions and constraints can be mathematically modeled in a conventional VRP formulation. However, a remaining challenge is the management of the mixed package delivery mechanism, where a customer's package may be picked up and delivered, respectively, by either a truck or a drone. As a result, it is impossible to apply the conventional precedence constraints to prevent the infeasible solution presented in Fig. 6.

Another challenge arises from the calculation of energy consumption. In order to calculate the total energy consumed during the flight, the total weight of the packages when launching should be precalculated. This calculation requires an accurate determination of the route of each package rather than a simple summation of all needs in a flight. Therefore, a number of auxiliary variables are introduced to record the route of each package, that is, to record whether a particular package is loaded on the truck or on the drone when it leaves a node.

Due to the introduction of a large number of variables and the complex constraints, modeling the whole problem with the methods of mixed integer linear programming is extremely complicated. It is difficult to obtain an optimal solution for such a complex model with conventional solution techniques. Therefore, the specific model is presented in Appendix A. The main function of this mathematical model was to verify the effect of the algorithm proposed in Section 4. The specific experimental results obtained for small-scale test cases are reported and discussed in Section 5.2.

4. Iterated local search metaheuristic for the problem

The iterated local search (ILS) metaheuristic applies a diversification mechanism and a local search (LS) procedure iteratively to search for better solutions. At each iteration, a new solution is generated by randomly performing a perturbation on a good local optimal solution found previously. The new perturbed solution is improved using the LS procedure. Through the perturbation mechanism, ILS retains part of the high-quality solution to generate promising initial solutions, and has been proven to be a simple yet successful approach for solving various transportation-related optimization problems (Ibarra-Rojas and Rios-Solis, 2012; Jorge et al., 2015; Zhang et al., 2015; Lim et al., 2017). A detailed explanation of the ILS metaheuristic can be found in Lourenço et al. (2019).

We designed a customized perturbation mechanism to construct different neighborhoods based on the distinctness of the problem. The mechanism automatically guides the selection of perturbing objects (either routes or requests) based on six specially-designed operators. Moreover, we integrated a series of local search operators for comprehensive exploitation to increase the probability of finding a local optimum. A general ILS framework is presented in Algorithm 1. In the framework, we first construct a new initial solution and pass it to the ILS procedure. The ILS procedure stops when either the maximum number of iterations or the maximum number of non-improved iterations has been met.

We introduce the solution representation and the feasibility evaluation method in Section 4.1. The two-phase method construction algorithm is presented in Section 4.2. The local search operators are described in detail in Section 4.3. Finally, the six perturbation operators are outlined in Section 4.4.

4.1. Solution representation and evaluation

A general solution with combined routes is represented by S $(S_1, S_2, \dots, S_k)$, where S_k represents the kth combined route including both the truck route GR and the drone trips UR. UR is further decomposed into $(UR_1, UR_2, \dots, UR_m)$, where UR_m represents the mth drone trip. Let $l = \{l_1, l_2, \dots, l_m\}$ and $r = \{r_1, r_2, \dots, r_m\}$ denote the set of

[3] The values of these parameters were set as $\lambda_1 = 2$ and $\lambda_2 = 1000$.

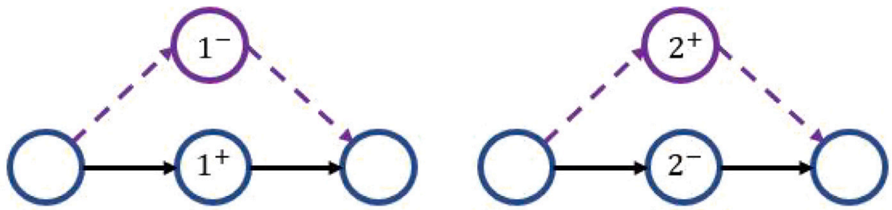


Fig. 6. Infeasible situation.

Algorithm 1 ILS

Input

Six perturbation operators including three route-based operators and three request-based operators

Construct a new initial solution S_0 (Section 4.2)
$S \leftarrow Localsearch(S_0)$
while Stop criteria is not met **do**
  $i \leftarrow 1$
  while $i \leq 6$ **do**
    $S' \leftarrow perturbation(S, i)$ (Section 4.4)
    $S'' \leftarrow Localsearch(S')$ (Section 4.3)
    if $obj(S'') < obj(S)$ **then**
      $S \leftarrow S''$
    end if
    $i \leftarrow i + 1$
  end while
end while

Output

Best found solution S

Table 1
Sample of solution presentation.

S	S_1	GR	$(0, 4, 5, 14, 3, 15, 13, 11, 21)$
		UR (UR_1, UR_2, UR_3)	$(\ (0, 2, 12, 14), (14, 1, 15), (15, 6, 16, 21)\ )$
		l (l_1, l_2, l_3)	$(0, 14, 15)$
		r (r_1, r_2, r_3)	$(14, 15, 21)$
	S_2	GR	$(0, 7, 17, 8, 19, 18, 21)$
		UR (UR_1, UR_2)	$(\ (0, 9, 17), (17, 10, 20, 18)\ )$
		l (l_1, l_2)	$(0, 17)$
		r (r_1, r_2)	$(17, 18)$

launch nodes and the set of retrieval nodes, respectively, where l_m and r_m indicate the launch node and the retrieval node of the mth drone trip, respectively. With reference to Fig. 3 in Section 1, the OPDP-MTMV solution is stored using a tailored data structure as shown in Table 1.

A feasible combined route for the OPDP-MTMV must fulfill all precedence constraints, energy consumption constraints, and capacity constraints for all trucks and drones. We break down the feasibility checking into three stages as summarized in Table 2.

4.1.1. Precedence constraint evaluation

Three types of hybrid service models exist for the OPDP-MTMV as shown in Fig. 4 in Section 1. As one combined route can be represented as a Directed Acyclic Graph (DAG), where the customers are represented as nodes, and the precedence orders of visits by the drone or the truck are represented by directed arcs, we apply a topological sort on the customer nodes of a combined route for the evaluation. The procedure can be evaluated in $O(|V|)$ time for a route with $|V|$ customer locations. The pseudo-code is provided in Algorithm C.1 in Supplemental Materials.

4.1.2. Capacity constraint evaluation for truck route

As packages may be picked up by the truck and delivered by the drone (and vice versa) and multiple pickup and delivery operations can be performed on a single drone trip, it is critical to evaluate the total weight carried by the drone and the truck at launch and retrieval, respectively, to enforce energy consumption and capacity constraints. Therefore, we introduce $Take_m$ to denote the total payload of the drone for the mth drone trip at launch node l_m, and $Drop_m$ to denote the total payload unloaded to the truck at retrieval node r_m. $Take_m$ and $Drop_m$ are calculated following Algorithm 2 for a given drone trip UR_m. If the drone delivers a package to the location of node i but does not pick up the package at the location of node $i - n$ in the same drone trip, the package must be taken from the launch node (Algorithm 2, Line 3–5). Similarly, if the drone picks up a package at the location of node i but does not deliver the package at the location of node $i + n$ in the same drone trip, the package must be dropped at the retrieval node (Algorithm 2, Line 6–8).

Algorithm 2 Loading information of launch and retrieve nodes

Input

m-th drone trip $UR_m(l_m, \ldots, r_m)$

1: Initial $Take_m \leftarrow 0$; $Drop_m \leftarrow 0$
2: **for** each node i in UR_m **do**
3:   **if** $i \in D,\ i - n \notin UR_m$ **then**
4:     $Take_m \leftarrow Take_m + w_{i-n}$
5:   **end if**
6:   **if** $i \in P,\ i + n \notin UR_m$ **then**
7:     $Drop_m \leftarrow Drop_m + w_i$
8:   **end if**
9: **end for**

Output

$Take_m$; $Drop_m$

The total weight carried by the truck changes when it serves a customer, launches a drone, or retrieves a drone, since the drone leaves with packages for delivery at the launch node and may be retrieved with some payload from one or more pickups. We denote the truck route as $GR(n_0, n_1, \ldots, n_q)$, where n_0 and n_q represent the depot, and denote the total weight carried by the truck when leaving customer n_i as $q^G_{n_i}$. Note that $q^G_{n_i}$ includes the self-weight of the drone if the drone is on board. When the truck departs from the depot, $q^G_{n_0} = w^U$ if the drone is not launched at the depot. Otherwise, $q^G_{n_0}$ is set to zero. The total weight $q^G_{n_i}$ of the truck when leaving customer n_i can then be calculated according to the following four scenarios:

$$q^G_{n_i} = \begin{cases} q^G_{n_{i-1}} + w_{n_i} & n_i \notin l \cup r \\ q^G_{n_{i-1}} - w^U + w_{n_i} - Take_m & n_i = l_m \in l \setminus r \\ q^G_{n_{i-1}} + w^U + Drop_m + w_{n_i} & n_i = r_m \in r \setminus l \\ q^G_{n_{i-1}} + w_{n_i} + Drop_{m-1} - Take_m & n_i = r_{m-1} = l_m \in l \cap r, \end{cases} \tag{1}$$

where $Take_m$ and $Drop_m$ represent the drone's total payload at launch and total payload of all unloaded packages after retrieval, respectively. At a launch node with a pickup operation, the peak of the total weight of the truck occurs after the package has been picked up. The total weight then decreases as the drone is loaded and departs from the truck. Conversely, at a retrieval node with a delivery operation, the peak occurs after the drone is retrieved and unloaded and before the

Table 2
Summary of feasibility checking.

Stage	Object	Constraints	Special attention
1	One combined route	Precedence	Parcels taken at launching nodes, and dropped at retrieving nodes
2	Truck route	Capacity	The changes in load at launching and retrieving nodes
3	Drone trips	Capacity, power	Extra payloads at launching nodes

package is delivered. Therefore, the maximum capacity constraint of the drone is verified against the peak as follows: a truck route is feasible at node n_i if $q^G_{n_i} \le Q^G$ for $n_i \notin l$, $q^G_{n_{i-1}} + w_{n_i} \le Q^G$ for $n_i = l_m \in l \setminus r$, and $q^G_{n_{i-1}} + w_{n_i} + Drop_{m-1} + w^U \le Q^G$ for $n_i = r_{m-1} = l_m \in l \cap r$.

4.1.3. Drone-level feasibility evaluation

A feasible drone trip needs to satisfy the following conditions: (1) The drone trip must finish before the energy consumption exceeds the drone's battery capacity; (2) The drone's total payload may not exceed the drone's maximum capacity; (3) For each request, the pickup node is visited by the truck or the drone before the delivery node.

For node n_i in the mth drone trip UR_m $(l_m, n_1, \dots, r_m)$, where l_m and r_m are the launch and retrieval nodes respectively, we calculate the total weight $q^U_{n_i}$ and remaining power $P^U_{n_i}$ when the drone leaves node n_i as follows:

$$q^U_{n_i} = \begin{cases} w^U + Take_m & n_i = l_m \in l \\ q^U_{n_{i-1}} + w_{n_i} & n_i \notin l \cup r \\ 0 & n_i = r_m \in r, \end{cases} \tag{2}$$

$$P^U_{n_i} = \begin{cases} \theta & n_i = l_m \in l \\ P^U_{n_{i-1}} - \alpha q^U_{n_{i-1}} t^U_{n_{i-1},n_i} - \alpha w^U_{n_{i-1}} s^U_{n_i} & n_i \notin l \cup r \\ P^U_{n_{i-1}} - \alpha q^U_{n_{i-1}} t^U_{n_{i-1},n_i} - \alpha q^U_{n_{i-1}} max(t^{U,A}_{n_i} - t^{G,A}_{n_i}, 0) & n_i = r_m \in r. \end{cases} \tag{3}$$

At the launch point ($n_i = l_m \in l$), the battery is fully charged and the total weight of the drone is equal to the sum of its self-weight and the total payload, while at the retrieval point ($n_i = r_m \in r$), the total payload of the drone is always set to 0, since the drone is on board and all the collected packages are unloaded onto the truck; the remaining energy is equal to the remaining energy at the last node of the trip ($P^U_{n_{i-1}}$) minus the energy consumed during the flight ($\alpha q^U_{n_{i-1}} t^U_{n_{i-1},n_i}$) and during the time spent hovering ($\alpha q^U_{n_{i-1}} max(t^{U,A}_{n_i} - t^{G,A}_{n_i}, 0)$), if any. Finally, when the drone visits and serves a node ($n_i \notin l \cup r$), the payload is updated based on w_{n_i}, and the drone consumes energy during both the flight time and the service time.

If $q^U_{n_i} \le Q^U + w^U$ and $P^U_{n_i} \ge 0$, both the capacity and energy consumption constraints are satisfied for the drone trip. A combined route is feasible only if the truck route and the drone trips are established as feasible. Our algorithm only accepts feasible solutions.

4.2. Constructive algorithm

Initial solutions are generated using a two-step approach. In the first step, we apply a truck-first-drone-second random algorithm to create solutions with combined routes. In the second step, we apply a greedy insertion to shift customers from the truck route to drone trips in order to reduce the duration costs and increase the drone utilization rates.

4.2.1. Creation of combined truck–drone routes

The truck-first-drone-second algorithm consists of a truck route construction stage and a drone trip construction stage, which process both the pickup node and the delivery node of a request at the same time. In the truck route construction stage, a request is randomly inserted into a feasible position in a truck route as long as the resultant truck route is feasible with respect to capacity constraints, precedence constraints and T_{max}. Afterwards, in the drone trip construction stage, the algorithm randomly picks a request and inserts its pickup node and delivery node into an existing or newly-created drone trip. The algorithm will create a new combined route when no request can be inserted feasibly into the current combined route.

4.2.2. Greedy truck-to-drone insertion

However, the truck-first-drone-second algorithm can create solutions using only trucks for test instances with small numbers of customers, which may lead to poorer performance of the local search operators (Section 4.3). In such a case, we apply a greedy insertion algorithm adopted from Schermer et al. (2018), which attempts to move nodes from the truck route to drone trips. Note that the insertion is based on a node instead of a request. Whenever necessary and feasible, the algorithm can also create a new drone trip for the node. Based on preliminary experiments, this approach is highly effective and necessary for the small-scale test instances we analyzed.

4.3. Local search

This section contains a discussion of the swap and relocate operators used in local search (Algorithm 1, Line 7). The designation considers the characteristics of truck–drone routes. Specifically, there are two general types of operators, the intra-route operator, that changes the node sequence in a combined route, and the inter-route operator, that interchanges two requests between two combined routes. Two intra-route operators, containing Intra-swap and Intra-relocate, are introduced in Section 4.3.1. One inter-route operator, Inter-swap operator, is introduced in Section 4.3.2.

4.3.1. Intra-route operators

According to the key features of the intra-swap and intra-relocate operations, four types of intra-swap operators and five types of intra-relocate operators are distinguished. These nine different types of operators are summarized in Table 3. These nine operators may change the structure of the combined route or the launch and retrieval nodes, which are grouped into the following three categories.

Category A is the most straightforward case with no changes to the structure of the combined route or the launch and retrieval nodes.

Category B operators update the launch and/or retrieval nodes, but the combined route structure remains unchanged. Such operators include Intra-swap types 2 (Fig. 7), 3 (Fig. 8), and 4 (Fig. 9), which simply swap the positions of two nodes and keep the combined route structure unchanged. Another such operator is Intra-relocate type 1, which maintains the relative positions of the launch and retrieval nodes in the combined route but changes the actual launch and retrieval nodes (from 2- to 4+) (Fig. 10). Algorithm 3 updates the launch and retrieval nodes after the local search operator has concluded, when the structure of the combined route is unaffected. The drone's launch and retrieval locations for the original combined route are reused for the newly-generated combined route.

Category C operators change both the structure of the combined route and the launch and retrieval nodes. When a node is moved from one drone trip to another drone trip (Intra-relocate type 2), the longer flight time resulting from the addition of an extra node to the trip will result in a longer waiting time for the truck, increasing its duration cost. Hence, the drone is launched earlier (Fig. 11). Fig. 12 (Intra-relocate type 3) and Fig. 13 (Intra-relocate type 4) illustrate the ordering of the affected drone trip when the node is shifted from the original segment. The retrieval node is advanced to the previous node visited by the truck before the original node and the launch node is postponed to the subsequent node in the truck route. This method

Table 3
Summary of intra local search operators. Column "Structure?" indicates whether the combined route structure is changed. Column "Launch/retrieval?" indicates whether the launch and/or retrieval nodes are updated. Column "New?" indicates whether a new drone trip is created.

Operation	No. of type	Domain	Node/Request	Category	Structure?	launch/retrieval?	New?
Intra-swap	1	A drone trip	Two nodes	A	N	N	N
	2	A truck route		B	N	Y	N
	3	A truck route and a drone trip		B	N	Y	N
	4	Combined route	Two requests	B	N	Y	N
Intra-relocate	1	Drone trip to truck route	One node	B	N	Y	N
	2	Two drone trips		C	Y	Y	N
	3	The truck route		C	Y	Y	N
	4	Truck route to drone trip		C	Y	Y	N
	5	Truck route to new drone trip		C	Y	Y	Y

Algorithm 3 Update launch and retrieval nodes of newly-generated combined route

Input
Original combined route S_k $(GR, UR\{UR_1, UR_2, \dots\})$
Newly-generated combined route $S_k^{'}$ $(GR^{'}, UR^{'}\{UR_1^{'}, UR_2^{'}, \dots\})$

```
for each drone route UR'_m in drone routes UR' do
    Find locations p, q such that GR[p] = l_m, GR[q] = r_m,
    where l_m, and r_m are the launch and retrieval nodes of UR_m
    Update l'_m ← GR'[p], r'_m ← GR'[q],
    where l'_m, and r'_m are the launch and retrieval nodes of UR'_m
end for
if S'_k is feasible then
    return S'_k
else
    return S_k
end if
```

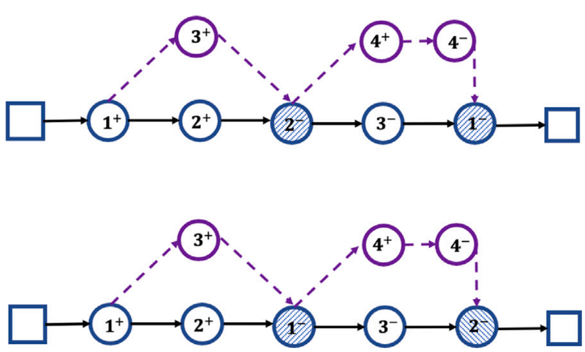


Fig. 7. Intra-swap type 2 within a truck route.

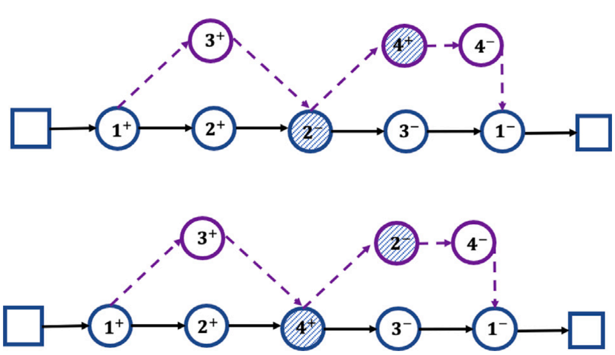


Fig. 8. Intra-swap type 3 between a truck route and a drone trip.

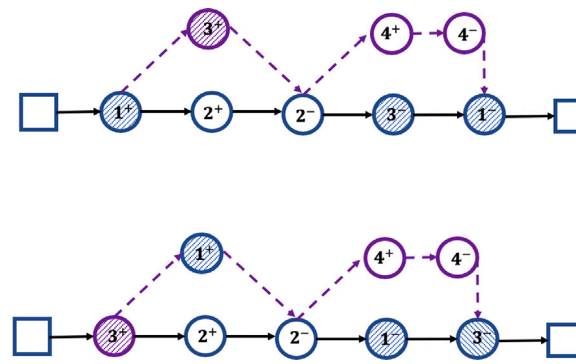


Fig. 9. Intra-swap type 4 with two requests.

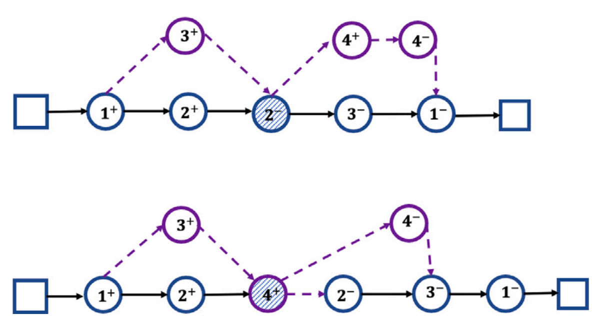


Fig. 10. Intra-relocate type 1: From drone to truck.

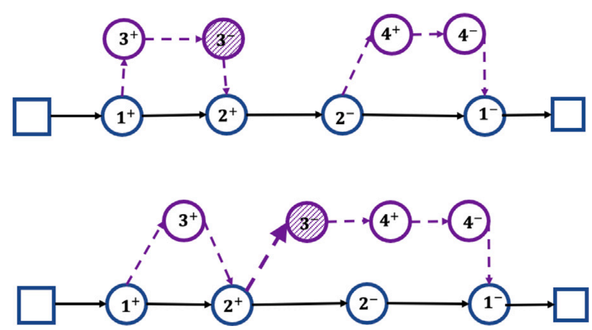


Fig. 11. Intra-relocate type 2: From one drone trip to another drone trip.

maintains the relative sequence of all subsequent nodes and potentially reduces energy consumption.

Based on preliminary experiments, it is beneficial to relocate a node from the truck to a new drone trip to increase the utilization rate of the drone, as it increases the likelihood of finding a better solution. The principle is similar to the cost-saving method in Clarke

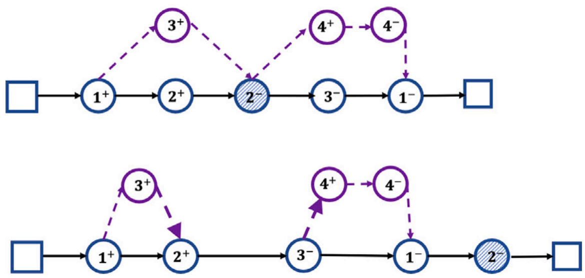


Fig. 12. Intra-relocate type 3: Within the truck route.

and Wright (1964). For this purpose, we introduced Intra-relocate type 5, which is designed to work on truck routes in which the first truck route segment contains no drone trip, that is, when the drone is not launched at the depot but is carried to the first customer node by the truck. First, we designed two scenarios based on the type and number of nodes selected. In the first scenario, both the pickup node and its corresponding delivery node are shifted to a new drone trip when both nodes appear in the first truck route segment. This simplifies the checking of the precedence constraint while reducing the total travel time required. In the second scenario, only the pickup node is shifted to a new drone trip if its corresponding delivery node is not in the same truck route segment. Next, the launch node is randomly selected from the set of possible nodes and the retrieval node is determined using a greedy best-cost-saving method.

4.3.2. Inter-swap operator

We applied the traditional VRP Inter-swap operator to exchange the positions of two requests from different combined routes with the best improvement strategy.

To accelerate the search process, we adopted the closeness measure in Shaw (1998) and the dynamic number of closed requests. The closeness measure of two requests i and j is defined as follows:

$$RL(i,j) = \gamma_1(d_{i,j} + d_{i+n,j+n}) + \gamma_2|w_i - w_j| \tag{4}$$

where γ_1 and γ_2 are normalized factors to be tuned. Let request i from route S_1 be evaluated for Inter-swap with another route S_2. First, the requests in S_2 are sorted in ascending order into a sorted list L based on the relative closeness measure with respect to i. Next, the maximum number of requests from S_2 for evaluation by Inter-swap is defined as $N = max\{1, \lceil \epsilon NumR \rceil\}$, where $\epsilon \in [0,1]$ is a pre-determined constant, and $NumR$ is the total number of requests visited in S_2. Lastly, the local search operator swaps the positions of request i and all requests j that are within the first n requests in the sorted list L.

4.4. Perturbation operators

Two mechanisms were designed for the problem of perturbing the current solution (Algorithm 1,Line 6), based on either requests or combined routes. Each mechanism contains three operators to select the requests, which are removed from their current positions and subsequently reinserted into new positions to obtain a new solution.

4.4.1. Request-based removal operator

The random request removal operator and the worst travel-duration removal operator remove one request at a time. The random request removal operator randomly selects one of the requests, while the worst travel-duration removal operator selects the request with the highest travel cost for removal. Formally, the marginal travel cost of request i is defined as the cost reduction achieved by removing both i and $i+n$ from the solution. This method increases the probability that the worst travel-duration removal operator will move a sub-optimally placed request to a different position in the solution. The random Shaw removal operator randomly removes one request and picks a second request based on the Shaw removal principle. The closeness measure is defined as in Eq. (4) for the Shaw removal operator.

Table 4
Summary of test instances.

Group	#Inst.	Characteristics	min ω_i	max ω_i	s_i	t_{max}	Q^G
lc	9	Cluster	10	50	90	1236	200
lr	12	Random	1	41	10	230	200
lrc	8	Cluster & Random	2	40	10	240	200

4.4.2. Route-based removal operator

We designed three operators to select a combined route whose requests will be removed from the solution. The random route removal operator selects a combined route randomly; the empty drone trip removal operator randomly selects one of the combined routes that contains no drone trips; the least efficient route removal operator selects the combined route(s) with the smallest number of served requests and makes a random selection if more than two combined routes share the same number of served requests in the routes.

4.4.3. Reinsertion

If the removal operator has removed two or more requests, the requests will be re-arranged randomly for ordered reinsertion. A greedy approach is applied to insert each request R_i $(i, i+n)$ into the best position with one of three hybrid service models, if possible. The selection of the service model is based on the following order: (1) Both requests are served by the drone; (2) One request is served by the drone and the other is served by the truck; (3) Both requests are served by the truck. This approach typically encourages greater drone utilization and improves the overall operational efficiency.

5. Computational experiments

In this section, we report on a series of experiments on the effectiveness of the OPDP-MTMV model and the performance of the proposed ILS algorithm. The programs were implemented in Python 3.8, and all the experiments were run on an Ubuntu 18.04.3 LTS server with a 2.10 GHz Intel(R) Xeon(R) CPU and 512 GB RAM. The ILS program was executed five times with different random seeds for each test run. The maximum number of iterations and the maximum number of non-improved iterations were set as 200 and 50, respectively. When either of these two stop criterion was met, the ILS was terminated.

5.1. OPDP-MTMV test instance

New test instances for the OPDP-MTMV were generated from the PDPTW test instances (Li and Lim, 2003). Test instances of various numbers of requests (n) were created for the experiments: (1) Small-scale instances with 3, 4, and 5 requests (Section 5.2); (2) Medium-scale instances with 25 requests (Section 5.3 and Section 5.4; (3) Large-scale instances with 50 requests (Section 5.5); (4) A group of instances with 10 requests specially for solution visualization in Section 5.6. The OPDP-MTMV test instances and generated solutions are available online at http://alim.algorithmexchange.com/orlib/MTDPDP-MV (see Table 4).

5.2. Performance analysis on small-scale instances with CPLEX

In this section, we evaluate the correctness and effectiveness of the ILS against those of the exact solver CPLEX 12.8. The maximum run time of CPLEX was set to 7200 s. The maximum truck capacity, drone self-weight, drone capacity, and battery capacity were set as $Q^G = 200$, $w^U = 5$, $Q^U = 50$, and $\theta = 800$. The travel speeds of the truck and drone are set to be 1 and 2, respectively.

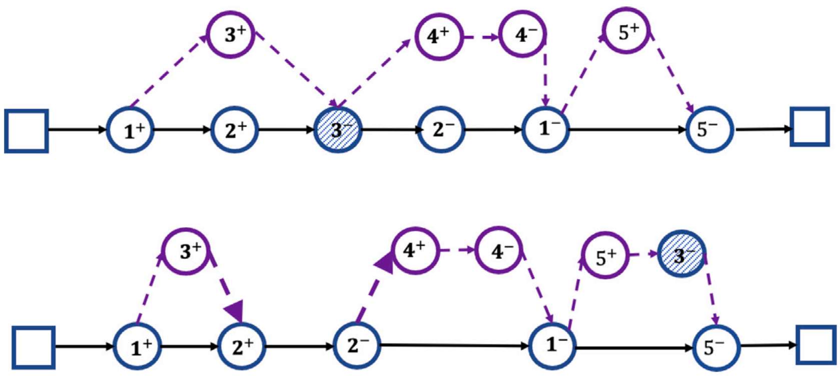


Fig. 13. Intra-relocate type 4: From truck to drone.

Table 5
Summary of results for small-scale instances.

Inst.	# inst	n = 3				n = 4				n = 5				
		CPLEX		ILS		CPLEX		ILS		CPLEX		ILS		
		# Opt	T_{CPLEX}	# C_{bst}	T_{ILS}	# Opt	T_{CPLEX}	# C_{bst}	T_{ILS}	# Opt	T_{tot}	# C_{bst}	T_{CPLEX}	Gap_{CPLEX}
lc	9	9	25.23	9	1.27	9	1196.77	9	6.11	0	7203.66	9	8.48	**−0.54%**
lr	12	12	32.03	12	0.88	12	1615.18	12	3.14 –	0	7200.30	12	3.46	**−5.58%**
lrc	8	8	31.44	8	1.39	8	1535.55	8	4.53	0	7203.82	8	5.46	**−1.27%**

For each instance group, Table 5 shows the number of optimal solutions obtained by the CPLEX solver ($\#Opt$), the number of similar or superior solutions obtained by the ILS ($\#C_{bst}$), and the average computational time required per group instance by the CPLEX solver (T_{CPLEX}) and by the ILS (T_{ILS}). In summary, both the CPLEX solver and the ILS found the exact optimal solutions of all test instances with 3 and 4 requests. However, the ILS used much less computational time than the CPLEX Solver.

For instances involving 5 requests, CPLEX could not solve any instances to optimality within the given runtime. The last column in Table 5, Gap_{CPLEX}, presents the average percentage gap, calculated as $(C_{bst} - UB)/UB$, where C_{bst} is the best objective value obtained by the ILS and UB is the best Upper Bound solution cost obtained by CPLEX. Note that the computational time required by CPLEX increases by a very large factor as the instance size increases, whereas the increase in time required by the ILS displays a more moderate trend. The detailed results are shown in Table D.1 and Table D.2 in Supplemental Materials.

5.3. Performance analysis on medium-scale instances

In Section 3, we introduced two parameters β_1 and β_2 to linearize the energy consumption function $f_3(t,w)$ in the model, so that commercial solvers can directly solve the model for small-scale test instances. We used the original form of $f_3(t,w) = \alpha \times w \times t$ as in Zhang et al. (2021), Liu et al. (2020) and Luo et al. (2021) to test the algorithm on instances with 25 requests. For evaluation, we defined a basic type drone with $Q^U = 25$ and $\theta = 2000$. The travel speeds of the truck and drone are kept 1 and 2. Three comparative experiments were designed to provide insight into OPDP-MTMV, including truck-only routes vs. truck–drone routes (Section 5.3.1), single-visit vs. multi-visit models (Section 5.3.2), and non-mixed vs. mixed pickup and delivery models (Section 5.3.3).

5.3.1. Comparison of truck-only routes to truck–drone routes

We restricted the use of drones in the solution and compared the results of truck-only solutions with the original truck–drone routes. The average objective value of truck–drone solutions is lower than that of the truck-only solutions. Furthermore, the average number of trucks used in the truck-only solutions is larger than in the truck–drone solutions, and the average cost of duration of truck-only solutions is higher

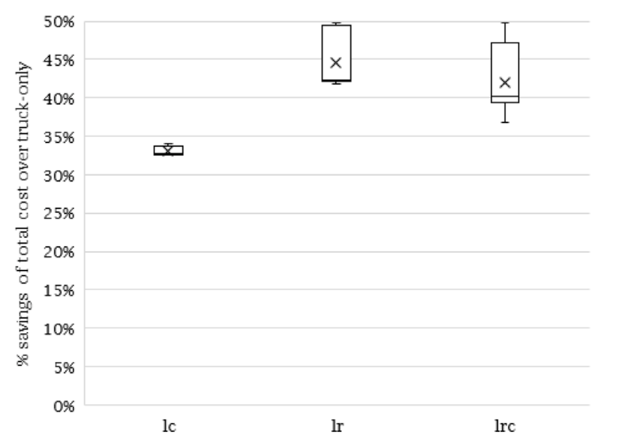


Fig. 14. Truck-only v.s. truck–drone, n = 25; box plot for lc, lr, lrc instance groups.

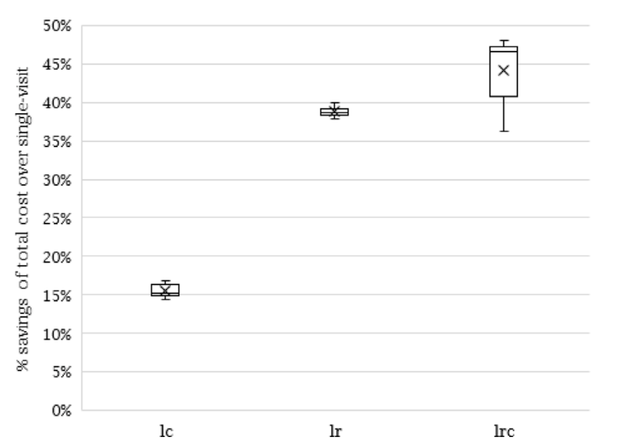


Fig. 15. Single-visit v.s. multi-visits, n = 25; box plot for lc, lr, lrc instance groups.

than that of truck–drone solutions. These observations demonstrate that cooperative interaction between a drone and a truck contributes to lower fixed costs and higher service efficiency. The average results of the best found solutions (BFS) using ILS based on different test instance groups are shown in Table D.3 in Supplemental Materials.

Box plots of the savings in total costs of the truck–drone solutions compared with the truck-only solutions are presented in Fig. 14, for different instance groups. The saved total costs of the solution exceeds 30% for each instance, most prominently for the instance of randomly distributed locations (lr), where the savings approached 50%. The box plot for the instance of cluster distributed locations (lc) corresponds to savings of approximately 33%, while the box plot for the instance of mixed distributed locations (lrc) displays the largest interquartile range, from 39.35% to 47.41%. These observations indicate that, from the perspective of reducing total costs, the advantage of interaction between a drone and a truck is especially pronounced in dispersed areas, even if the variance of the cost savings for dispersed areas is marginally larger than for clustered areas. The detailed results are shown in Table D.4 in Supplemental Materials.

5.3.2. Comparison of single-visit to multi-visits

In order to study the impacts of multiple drone visits in the OPDP-MTMV, we ran two versions of the ILS on instances with 25 requests, in either multi-visits or single-visit mode. The average objective values and the average number of used trucks of the BFS for each instance group were calculated. We also introduced $Ratio$ as the preference of the algorithm in assigning customers to drones or trucks, which is calculated as ***the number of customers served by drones/the number of customers served by trucks***. The summarized results of the solutions are shown in Table D.5 in Supplemental Materials, where the average objective values, the average number of used trucks of the BFS, and $Ratio$ are included.

The lower average objective values and positive percentage saving shown in Fig. 15 demonstrates that allowing multiple drone visits leads to substantial cost savings for the OPDP-MTMV. In addition, Fig. 15 shows that the percentage of savings is related to the distribution of customers and service time. For the lc group with a long service time, the average cost savings of solutions with multi-visits is approximately 15%, which is much lower than that of solutions for the lr and lrc groups with their shorter service times, which exceeds 35%. In addition, the box plots of the savings for the lc and lr groups show a smaller variance than that of lrc group because the customer distribution of the lrc group is not uniform with that of the cluster and random groups. The average savings achieved for solutions of the lrc group (44.19%) is slightly higher than that for the lr group (38.75%). One possible explanation is that allowing multi-visits is greatly beneficial to the cluster customer distribution which has short service times.

Furthermore, the average number of trucks used is reduced by a factor of almost 2 when multi-visits are introduced, leading directly to a reduction of the fixed costs. In the multi-visit mode, more customers are assigned to individual drones, resulting in lower costs. This is intuitively reasonable because a drone, which moves faster than a truck, has the potential to serve more customers within a certain time period, thereby reducing the number of trucks used. This is confirmed by the lower value of *No. of trucks* and the corresponding increase in $Ratio$ in Table D.5 in Supplemental Materials. The detailed results are shown in Table D.6 in Supplemental Materials.

5.3.3. Comparison of a restrictive OPDP-MTMV with the standard OPDP-MTMV

We propose a restrictive OPDP-MTMV by requiring that: (1) The package picked up by a drone must be delivered by that drone; (2) A package picked up by a truck must be delivered by that truck; (3) A package picked up by a drone must be delivered immediately and directly in a single drone trip. In this section, we present the results of our investigation of the impact of allowing mixed pickup and delivery in the OPDP-MTMV vs. the performance of the restrictive OPDP-MTMV. The characteristics of the BFS of the restrictive OPDP-MTMV and OPDP-MTMV, respectively, are summarized in Table 6, in the form of the average objective value, cost of duration, number of drone trips per truck, and *Ratio* of the BFS for different instance groups.

As expected, the average objective value and the cost of duration of the BFS of the restrictive OPDP-MTMV are higher than those of the OPDP-MTMV for all instance groups. This demonstrates that the OPDP-MTMV is more conducive to saving costs than the restrictive OPDP-MTMV, including the fixed cost and the duration cost. It is noteworthy that for the lc group, the average number of drone trips per truck of the BFS of the OPDP-MTMV (3.39) is higher than for the restrictive OPDP-MTMV (3.19). This is the converse of what is observed for the lr and lrc groups.

This observation for the lc group may be attributed to the heavier packages and longer service times of the lc group, resulting in a higher power consumption. Consequently, the settings of the restrictive OPDP-MTMV, especially the restriction that packages must be directly delivered after being picked up, in a single drone trip, impose limitations on the number of drone trips that can be accommodated. Another possible reason for this observation is that serving one request with different vehicles (the truck and the drone) saves time when the distance between pickup and delivery locations for a particular customer is relatively great. Let us review the example solution in Fig. 19(d). As an example, we consider request 9. Drone 2 picks up the package at the location 9+ on its first drone trip. The package is delivered by Truck 1 to the location 9-. When Drone 2 picks up package 9, the remaining power 9 is not sufficient to allow the drone to deliver package 9 in a direct flight because of the long distance. Consequently, this package has to be delivered by Truck 1.

Furthermore, the values of the *Ratio* shown in Table 6 reveal that the number of customers assigned to the truck by the algorithm in the restrictive OPDP-MTMV is more than double the number assigned in the OPDP-MTMV. This observation implies that the restrictive settings cause some of the advantages of multiple drone visits be lost, thereby increasing the total costs.

To sum up, allowing the truck and its corresponding drone to perform pickup and delivery with the degree of flexibility inherent in the OPDP-MTMV provides solutions for long-distance service challenges in realistic situations. The OPDP-MTMV provides dispatchers with more choices in designing a truck–drone pickup and delivery routing plan. The detailed results are shown in Table D.7 in Supplemental Materials.

5.4. Impacts of the load and battery capacities of the drone on the performance of OPDP-MTMV

We run the ILS on instances with 25 requests and a basic drone with a load capacity Q^U varying from 15 to 55 and a battery capacity θ varying from 1000 to 5000. Note that all results represent average values of all 29 instances, including lc, lr, and lrc. The discussions based on the results mainly focus on the objective value of the BFS (Section 5.4.1) and the vehicle utilization (Section 5.4.2) of the obtained solutions. The detailed results are shown in Table D.8 in Supplemental Materials.

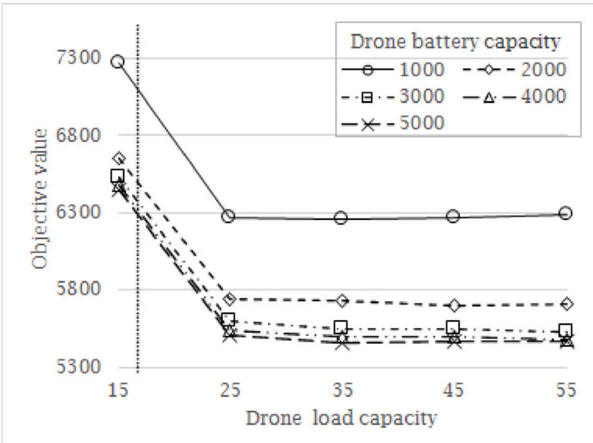
5.4.1. Impacts of drone load and battery capacity on the objective value of the BFS

The load capacity and its corresponding objective value of the BFS under different battery capacities are displayed in Fig. 16. The vertical dashed line in Fig. 16(a) represents the average weight of all packages, which has a value of 16.

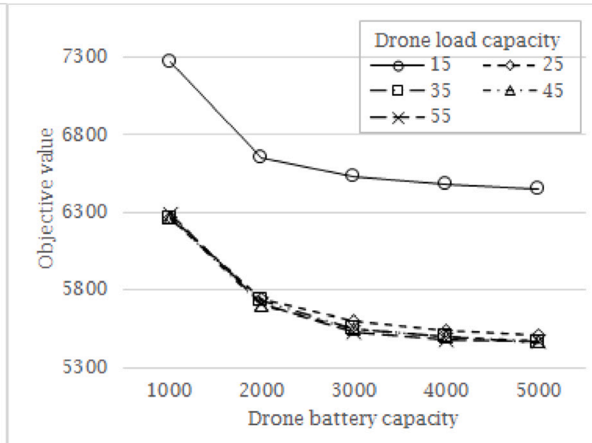
For each of the fixed drone battery capacities, the objective value of the BFS steeply decreased by approximately 14% when the drone load capacity increased from 15 to 25, as shown in Fig. 16(a). There was almost no subsequent change in the objective value of the BFS as the drone load capacity further increased up to a value of 55. Furthermore, the objective value of the BFS decreased with increasing drone battery capacity, indicating that a large drone battery capacity was preferable with respect to achieving more optimal objective values. However, beyond a level of 3000, further increases in the drone battery capacity

Table 6
Comparison of the restrictive OPDP-MTMV with standard OPDP-MTMV, $n = 25$, averaged over different instance groups.

Inst.	*Objective value*		*Cost of duration*		*No. of drone trips per truck route*		*Ratio*	
	Restrictive OPDP-MTMV	OPDP-MTMV	Restrictive OPDP-MTMV	OPDP-MTMV	Restrictive OPDP-MTMV	OPDP-MTMV	Restrictive OPDP-MTMV	OPDP-MTMV
lc	9202.00	8919.02	6202.00	5919.02	3.19	3.93	0.19	0.69
lr	4406.39	4144.29	1239.72	1144.29	2.58	1.86	0.58	1.80
lrc	4982.10	4376.62	1357.10	1251.62	2.27	2.00	0.63	1.60
Avg.	6196.83	5813.31	2932.94	2771.64	2.68	2.60	0.47	1.36



(a) Comparison of different drone load capacities



(b) Comparison of different drone battery capacities

Fig. 16. Impacts of drone load and battery capacities on the objective value of the BFS, averaged over all instances with $n = 25$.

delivered only marginal improvements in the objective value of the BFS.

A similar trend is observed in Fig. 16(b): For each fixed value of the drone load capacity, the objective value of the BFS decreased with increasing battery capacity of the drone. The initial decrease in the objective value of the BFS was steep, subsequently leveling out as the battery capacity was increased further. In addition, once the drone capacity had reached a value of 25, further increases in the load capacity led to very marginal improvements in the objective value of the BFS.

The above results demonstrate that drones with larger payload and battery capacities improve the performance measures of the truck–drone one-to-one pickup and delivery service. However, considering the extra costs ensuing from increasing these capacities, such as charging costs, production costs, and technology costs in relation to the marginal improvements observed beyond certain threshold values of load and battery capacities, the use of overcapable drones might not always be optimal. Judicious decisions regarding the load and battery capacities of a drone are required. For example, drone load and battery capacity thresholds of 25 and 2000, respectively, would constitute reasonable design parameters for the instances investigated in the aforementioned discussion, where the weight of each package ranges between 0 and 50, and the average weight is 16.

5.4.2. Impacts of drone load and battery capacity on the vehicle utilization of the BFS

The average number of customers served on one truck route and the average number of customers served on one drone trip are crucial characteristics of the truck–drone route, providing deeper insight into the vehicle utilization of the BFS. We also analyzed how the value of *Ratio* changed in response to changes in the drone load and battery capacities, illuminating the relationship between the number of customers served by the truck and the drone.

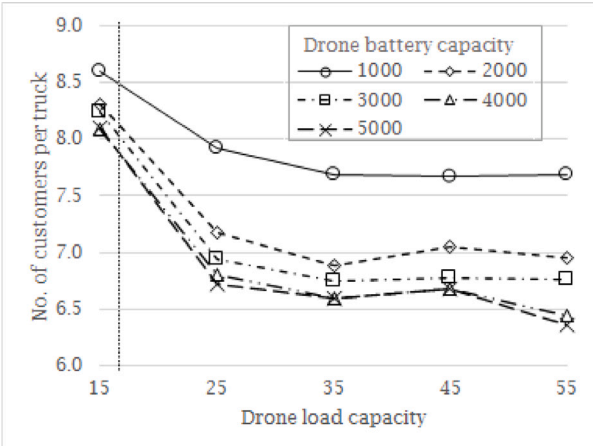
Fig. 17(a) shows that an increase in drone load capacity led to a decrease in the number of customers per truck. However, the extent of the decrease was restricted by the battery capacity. Once the drone capacity had increased to a certain level (approximately 25 in this study), greater gains were achieved by increasing the drone battery capacity than by increasing the drone load capacity. These trends are also illustrated in Fig. 17(b), which displays the relationship between drone load capacity and the number of customers visited in one drone trip of the BFS. The number of customers visited in one drone trip of the BFS increased with an increased load capacity of the drone. The converse gradients of the lines in Figs. 17(a) and 17(b) reflect the numerical imperative that the number of visited customers per truck and per drone trip should sum up to the same total in each instance.

The plot of the *Ratio* in Fig. 17(c) indicates that when the drone load capacity became smaller than the average weight of one package (a value of 15 — shown as the vertical dashed line), the value of *Ratio* fell below 1 for each of the lines associated with the different fixed battery capacities. A value of the *Ratio* below 1 indicates that most customers are assigned to the truck because the weights of most of the packages exceed the drone load capacity. A substantial increase in the *Ratio* was observed as the drone load capacity increased from 15 to 25. Beyond the value of 25, the value of the *Ratio* fluctuated slightly about a constant value for a battery capacity of 1000, whereas the value of the *Ratio* continued to increase with increasing load capacity for the larger values of battery capacity, albeit with some fluctuations.

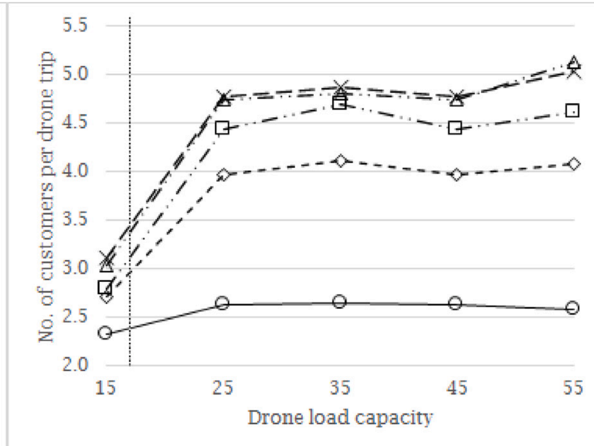
In summary, the vehicle utilization, as measured by the average number of customers per truck and per drone trip, respectively, was affected by both drone load and battery capacities.

5.5. Performance analysis on large-scale instances

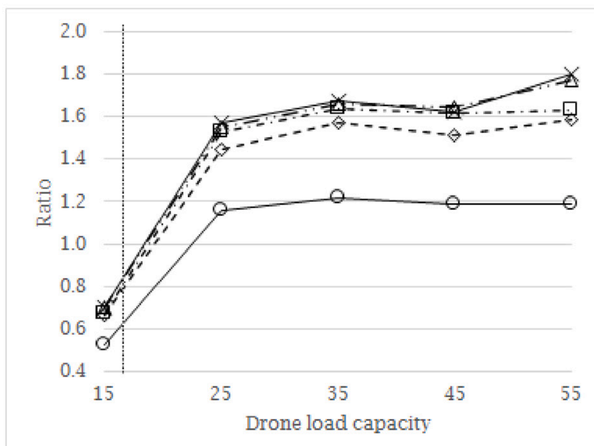
The design of the ILS includes a degree of randomness, for instance in the truck-first-drone-second random algorithm (Section 4.2), the Intra-route operators (Section 4.3.1), and the perturbation operators (Section 4.4). Consequently, different runs of the algorithm could generate different solutions. If the variance of the solutions obtained among different runs is large, the stability of the heuristic is affected. We discuss our examination of the heuristic stability for 50 requests in this sub-section.



(a) Impact on the number of customers per truck



(b) Impact on the number of customers per drone trip



(c) Impact on *Ratio*

Fig. 17. Impacts of drone load and battery capacities on vehicle utilization of solutions, averaged over all instances with $n = 25$.

The heuristic results related to the objective value are depicted in Fig. 18. For each instance, the gap between the objective value of the BFS (Best) and the average objective value of the solutions across five runs (Average) on the same instance is described as Gap = (Average-Best)/Best. The value of Gap of each instance in the lc group (leftmost in the plot) was below 1.5%. In contrast with the lc group, the values of Gap for the lr and lrc groups included two test instances for each group that had relatively high values of Gap. These were lr102 (8.75%), lr111 (3.22%), lrc104 (8.35%), and lrc 107 (10.76%). The values of Gap for the remaining instances of the lr and lrc groups were below 1.5%. The average value of Gap for all instances equaled 1.66%. These low values of Gap imply that the ILS is reasonably stable, whereas the random distribution of customer locations possibly generated a negative effect. The detailed results are shown in Table D.9 in Supplemental Materials.

5.6. Heuristic solution analysis

In this section, we discuss the results of running the ILS on instances with 10 requests. We display the resulting behavior of the BFS. The BFS exhibited different characteristics because of the varied customer distributions and service times of the test instances, as shown in Fig. 19. We make the following observations:

- It is reasonable that more trucks and drones are deployed when the distances between the depot and the customers are further (Figs. 19(a) and 19(b)).
- Trucks and drones are more likely to work independently of each other when the customers are randomly distributed (Fig. 19(a)). Conversely, trucks and drones tend to work in a cooperative way when the customers are distributed in clusters (Fig. 19(b).
- When the distances between the depot and the customers are relatively close, the number of trucks deployed is smaller, and the drones and trucks tend to work more cooperatively (Figs. 19(c) and 19(d)).
- As the endurance of drones depends on the payload and flight time, when the service time is long and the distances between customers and the depot are relatively further, the drone is more likely to fly in sidekick mode with the truck (Figs. 19(b) and 19(d)).
- The length-of-service time affects the assignment of customers significantly. Specifically, more customers will be assigned to and served by drones with a shorter service time.

6. Conclusion

The cooperative truck–drone deployment model holds substantial potential for improving the operational efficiency of pickup and delivery scenarios in commercial applications, for example in logistics and food delivery. In order to explore this potential, we introduced the novel OPDP-MTMV to improve and enhance existing truck–drone transportation models for greater realism and applicability.

The OPDP-MTMV has unique problem characteristics, including a hybrid truck–drone deployment, precedence requirements for pickup

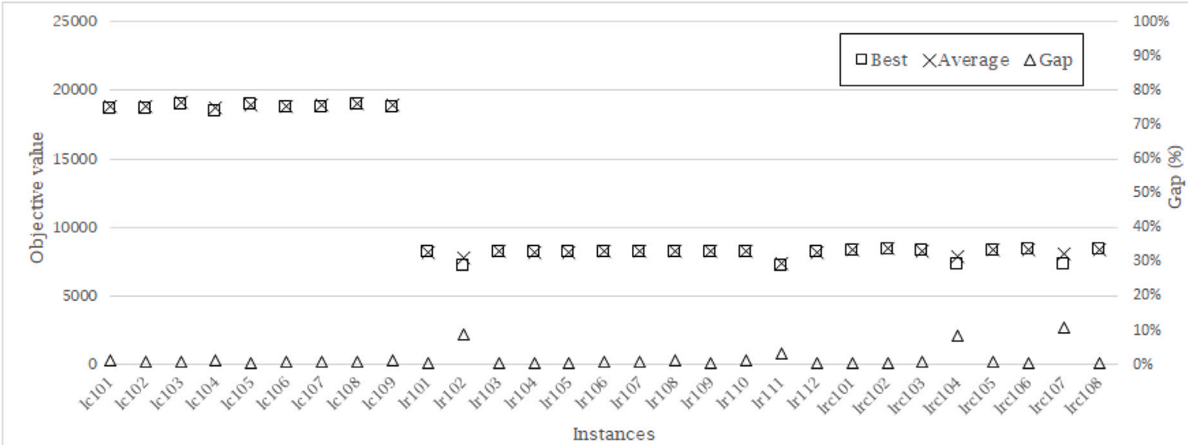
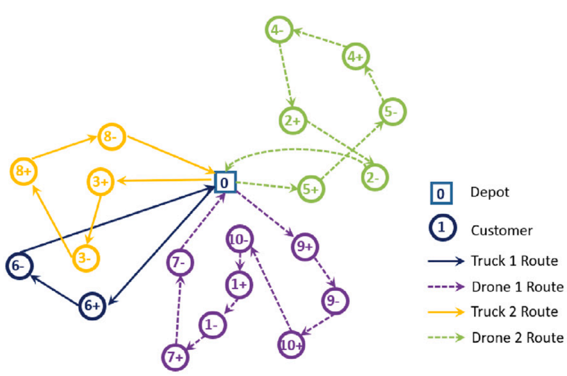
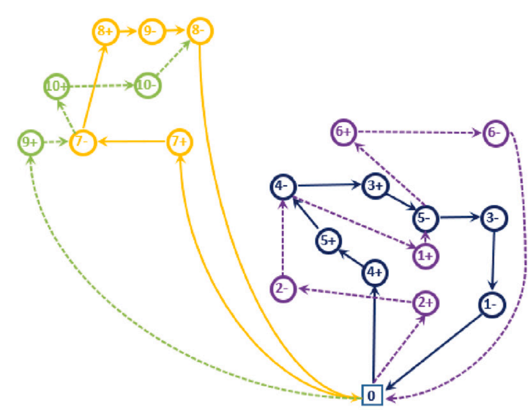


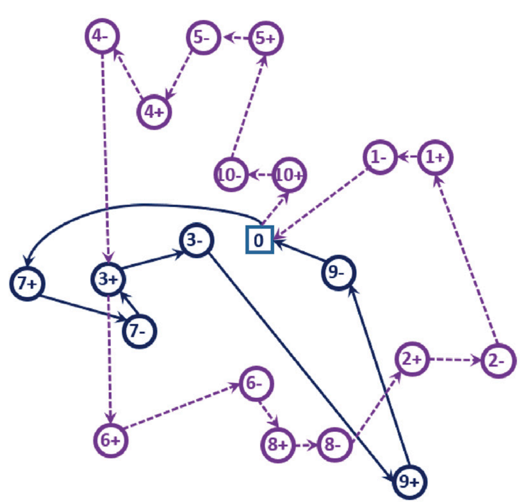
Fig. 18. Best, Average and Gap.



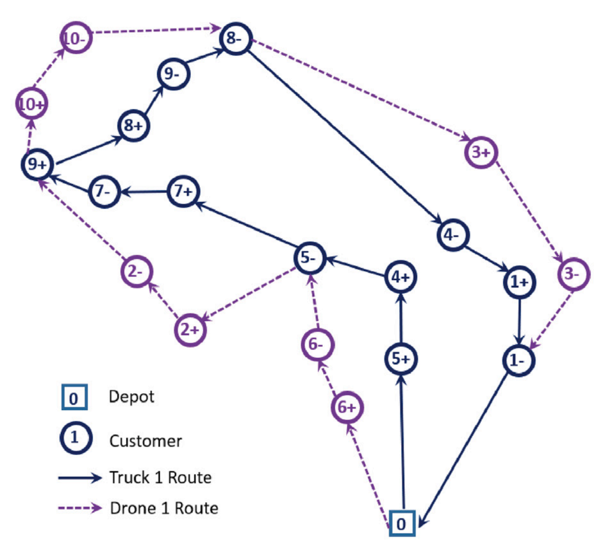
(a) lr105: Random-far distribution with short service time



(b) lc106: Cluster-far distribution with long service time



(c) lr102: Random-near distribution with short service time



(d) lc108: Cluster-near distribution with long service time

Fig. 19. Four kinds of typical solutions for 10 requests.

and delivery, and energy consumption during a payload-bearing drone's flight and hovering time. Our contributions in this work include the following: (1) Modeling the problem by decomposing truck–drone routes into segments; (2) Customizing the drone endurance model to consider the flight time, hovering time at retrieval, and payload; (3) Developing an iterated local search (ILS) metaheuristic with a three-stage feasibility evaluation method that utilizes tailored local search and perturbation operators; (4) Conducting comprehensive experiments to validate the correctness of the formulation and the effectiveness of the model.

Potential extensions to this work for future research could include more realistic modeling of the drone's endurance model, investigating the relevance of the model in real-world deployments, exploring better heuristic and exact techniques to solve the problem, and including additional requirements such as last-in-first-out constraints for packages to be delivered or picked up by drones.

CRediT authorship contribution statement

Zhihao Luo: Conceptualization, Methodology, Software, Investigation, Writing – original draft, Review & editing. **Ruixue Gu:** Methodology, Software, Validation, Investigation, Writing – original draft. **Mark Poon:** Conceptualization, Methodology, Validation, Investigation, Writing – review & editing, Supervision. **Zhong Liu:** Supervision, Project administration, Funding acquisition. **Andrew Lim:** Supervision, Project administration, Funding acquisition.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.cor.2022.106015.

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